

Progress Report — Week 10

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Reporting period

3 July 2026 – 9 July 2026

Hours worked

Date	Day	Hours	Activity summary
3 Jul 2026	Fri	8	Externalized weights to config/weights.yaml
6 Jul 2026	Mon	8	config_loader.py and model_version tracking
7 Jul 2026	Tue	8	Preset scenarios (high-trust, over-trust, low-CI, etc.)
8 Jul 2026	Wed	8	CSV/JSON export, extra UI charts (viability, sustainability, CI subcomponents)
9 Jul 2026	Thu	8	Assumptions panel in Streamlit; README v2.0 update
Total		40	

Work completed

- Released model v2.0 tooling: all CI and effectiveness weights moved to config/weights.yaml.
- Implemented config_loader.py for versioned, inspectable configuration.
- Added preset scenarios to Streamlit sidebar for common research demonstrations.
- Extended UI with charts for viability, sustainability, overload, and CI subcomponent breakdown.

- Added CSV/JSON export for single runs and experiment summaries.
 - Created in-app Assumptions tab showing formulas and externalized weights.
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Connection to learning objectives

- Applied software configuration and UX design principles to improve research tool usability.
 - Worked systematically toward milestone v2.0 with clear module boundaries.
 - Produced employer-visible value: a more transparent and demonstrable prototype.
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Challenges and how they were handled

Caching in Streamlit caused stale results after weight changes. I added cache invalidation on config reload and documented the behaviour in README.

Plan for next week (10–16 Jul 2026)

- Mixed-backlog generator with task-type proportions and carry-over reporting.
 - Task dependencies, blocker propagation, defect rework spillover.
 - Sprint-phase modifiers and lightweight Scrum roles.
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Student signature: Arslan Sultan — Date: 10 July 2026

Supervisor acknowledgment: Juan Garbajosa